

An example of representation in FOL (cont'd)

Example (Formulas)

Intended meaning	Formula
Aldo and Bruno are the same person	$Aldo = Bruno$
Carlo is a person and Math is a course	$person(Carlo) \wedge course(Math)$
Aldo attends Math	$attend(Aldo, Math)$
Courses are attended only by students	$\forall x \forall y (attend(x, y) \rightarrow course(y) \rightarrow student(x))$
every course is attended by somebody	$\forall x (course(x) \rightarrow \exists y attend(y, x))$
every student attends something	$\forall x (student(x) \rightarrow \exists y attend(x, y))$
there is a student who attends all the courses	$\exists x (student(x) \wedge \forall y (course(y) \rightarrow attend(x, y)))$
every course has at least two attenders	$\forall x (course(x) \rightarrow \exists y \exists z (attend(y, x) \wedge attend(z, x) \wedge \neg y = z))$
Aldo's best friend attend all the courses attended by Aldo	$\forall x (attend(Aldo, x) \rightarrow attend(best-friend(Aldo), x))$
Aldo and his best friend have the same mark in Math	$mark(best-friend(Aldo), Math) = mark(Aldo, Math)$
A student can attend at most two courses	$\forall x \forall y \forall z \forall w (attend(x, y) \wedge attend(x, z) \wedge attend(x, w) \rightarrow (y = z \vee z = w \vee y = w))$

Common Mistakes

- Use of $\wedge$ with $\forall$

$\forall x (WorksAt(FBK, x) \wedge Smart(x))$ means “Everyone works at FBK and everyone is smart”

“Everyone working at FBK is smart” is formalized as $\forall x (WorksAt(FBK, x) \rightarrow Smart(x))$

- Use of $\rightarrow$ with $\exists$

$\exists x (WorksAt(FBK, x) \rightarrow Smart(x))$ means “There is a person so that if (s)he works at FBK then (s)he is smart” and this is true as soon as there is at last an x who does not work at FBK

“There is an FBK-working smart person” is formalized as

$\exists x (WorksAt(FBK, x) \wedge Smart(x))$

Representing variations quantifiers in FOL

Example

Represent the statement “at least 2 students attend the KR course”

$$\exists x_1 \exists x_2 (attend(x_1, KR) \wedge attend(x_2, KR))$$

The above representation is not enough, as x_1 and x_2 are variable and they could denote the same individual, we have to guarantee the fact that x_1 and x_2 denote different person. The correct formalization is:

$$\exists x_1 \exists x_2 (attend(x_1, KR) \wedge attend(x_2, KR) \wedge x_1 \neq x_2)$$

At least n ...

$$\exists x_1 \dots x_n \left(\bigwedge_{i=1}^n \phi(x_i) \wedge \bigwedge_{i \neq j=1}^n x_i \neq x_j \right)$$

Representing variations of quantifiers in FOL

Example

Represent the statement “**at most 2** students attend the KR course”

$$\forall x_1 \forall x_2 \forall x_3 (attend(x_1, KR) \wedge attend(x_2, KR) \wedge attend(x_3, KR) \rightarrow x_1 = x_2 \vee x_2 = x_3 \vee x_1 = x_3)$$

At most n ...

$$\forall x_1 \dots x_{n+1} \left(\bigwedge_{i=1}^{n+1} \phi(x_i) \rightarrow \bigvee_{i \neq j=1}^{n+1} x_i = x_j \right)$$

Models, satisfiability and validity

In the following, if Γ is a set of formulas, then $(\mathcal{I}, \alpha) \models \Gamma$ means that $(\mathcal{I}, \alpha) \models g$ for all $g \in \Gamma$.

Definition (Model, satisfiability and validity)

An interpretation $\mathcal{I}$ is a **model** of ϕ under the assignment α , if $(\mathcal{I}, \alpha) \models \phi$.

A formula ϕ is **satisfiable** if there is some $\mathcal{I}$ and some assignment α such that $(\mathcal{I}, \alpha) \models \phi$.

A formula ϕ is **unsatisfiable** if it is not satisfiable.

A formula ϕ is **valid** if for every $\mathcal{I}$ and every assignment α , we have $(\mathcal{I}, \alpha) \models \phi$.

Definition (Logical implication)

A formula ϕ is **logical implied** by a set of formulas Γ , in symbols $\Gamma \models \phi$, if for all interpretations $\mathcal{I}$ and for all assignment α , if $(\mathcal{I}, \alpha) \models \Gamma$ then $(\mathcal{I}, \alpha) \models \phi$.

Models, satisfiability and validity for closed formulas

Definition (Model, satisfiability and validity for closed formulas)

An interpretation $\mathcal{I}$ is a **model** of a closed formula ϕ if $\mathcal{I} \models \phi$.

A closed formula ϕ is **satisfiable** if there is some $\mathcal{I}$ such that $\mathcal{I} \models \phi$.

A closed formula ϕ is **unsatisfiable** if it is not satisfiable.

A closed formula ϕ is **valid** if for every $\mathcal{I}$ we have $\mathcal{I} \models \phi$.

Definition (Logical implication)

A closed formula ϕ is **logically implied** by a set of closed formulas Γ , in symbols $\Gamma \models \phi$, if for all interpretations $\mathcal{I}$, if $\mathcal{I} \models \Gamma$ then $\mathcal{I} \models \phi$.

The notions of **theory**, **set of axioms of a theory**, and **finitely axiomatizable theory** in first-order logic is a straightforward generalization of the corresponding notions in propositional logic.

Properties of first-order logical implication

Proposition

If Γ and Σ are two sets of closed formulas formulas, and A and B two closed formulas, then the following properties hold:

Reflexivity If $A \in \Gamma$, then $\Gamma \models A$

Ex falso sequitur quodlibet If Γ is unsatisfiable, then $\Gamma \models A$ for all A

Monotonicity If $\Gamma \models A$ then $\Gamma \cup \Sigma \models A$

Cut If $\Gamma \models A$ and $\Sigma \cup \{A\} \models B$ then $\Gamma \cup \Sigma \models B$

Compactness If $\Gamma \models A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$, such that $\Gamma_0 \models A$

Deduction theorem If $\Gamma \cup \{A\} \models B$ then $\Gamma \models A \rightarrow B$

Deduction principle $\Gamma \cup \{A\} \models B$ if and only if $\Gamma \models A \rightarrow B$

Refutation principle $\Gamma \models A$ if and only if $\Gamma \cup \{\neg A\}$ is unsatisfiable

Computational properties of first-order logic

Theorem

Checking if a formula in first-order logic is valid is an undecidable problem (although semi-decidable: there is algorithm that terminates and answer “yes” if the formula is valid). The same holds for checking satisfiability and checking logical implication.

The proof is by reduction from the **halting problem**, that is the problem of checking if a given Turing machine halts for a given input. The basic idea is that **the Turing machine halting problem can be formalized by a first-order logic that is valid if and only if the machine halts for the input.**

What about the evaluation of a formula in an interpretation? Note that this question is meaningful only if the interpretation can be represented as a finite structure. Actually, it can be shown that **evaluating a formula in a finite interpretation is a PSPACE-complete problem.**

Exercises

Tell whether these formulas are valid, satisfiable, or unsatisfiable

- $\forall x P(x)$
- $\forall x P(x) \rightarrow \exists y P(y)$
- $\forall x \forall y (P(x) \rightarrow P(y))$
- $P(x) \rightarrow \exists y P(y)$
- $P(x) \vee \neg P(y)$
- $P(x) \wedge \neg P(y)$
- $P(x) \rightarrow \forall x P(x)$
- $\forall x \exists y Q(x, y) \rightarrow \exists y \forall x Q(x, y)$
- $x = x$
- $\forall x P(x) \equiv \forall y . P(y)$
- $x = y \rightarrow \forall x . P(x) \equiv \forall y P(y)$
- $x = y \rightarrow (P(x) \equiv P(y))$
- $P(x) \equiv P(y) \rightarrow x = y$

Solution

$\forall x P(x)$	Satisfiable
$\forall x P(x) \rightarrow \exists y P(y)$	Valid
$\forall x \forall y (P(x) \rightarrow P(y))$	Satisfiable
$P(x) \rightarrow \exists y P(y)$	Valid
$P(x) \vee \neg P(y)$	Satisfiable
$P(x) \wedge \neg P(y)$	Satisfiable
$P(x) \rightarrow \forall x P(x)$	Satisfiable
$\forall x \exists y Q(x, y) \rightarrow \exists y \forall x Q(x, y)$	Satisfiable
$x = x$	Valid
$\forall x P(x) \equiv \forall y P(y)$	Valid
$x = y \rightarrow \forall x P(x) \equiv \forall y P(y)$	Valid
$x = y \rightarrow (P(x) \equiv P(y))$	Valid
$P(x) \equiv P(y) \rightarrow x = y$	Satisfiable

Properties of quantifiers

Proposition

The following formulas are valid

- $\forall x(\phi(x) \wedge \psi(x)) \equiv \forall x\phi(x) \wedge \forall x\psi(x)$
- $\exists x(\phi(x) \vee \psi(x)) \equiv \exists x\phi(x) \vee \exists x\psi(x)$
- $\forall x\phi(x) \equiv \neg\exists x\neg\phi(x)$
- $\forall x\exists x\phi(x) \equiv \exists x\phi(x)$
- $\exists x\forall x\phi(x) \equiv \forall x\phi(x)$

Proposition

The following formulas are not valid

- $\forall x(\phi(x) \vee \psi(x)) \equiv \forall x\phi(x) \vee \forall x\psi(x)$
- $\exists x(\phi(x) \wedge \psi(x)) \equiv \exists x\phi(x) \wedge \exists x\psi(x)$
- $\forall x\phi(x) \equiv \exists x\phi(x)$
- $\forall x\exists y\phi(x, y) \equiv \exists y\forall x\phi(x, y)$

Expressing properties in FOL

What is the meaning of the following FOL formulas?

- ① $\exists x(bought(Frank, x) \wedge dvd(x))$
 - ② $\exists x bought(Frank, x)$
 - ③ $\forall x(bought(Frank, x) \rightarrow bought(Susan, x))$
 - ④ $(\forall x bought(Frank, x)) \rightarrow (\forall x bought(Susan, x))$
 - ⑤ $\forall x \exists y bought(x, y)$
 - ⑥ $\exists x \forall y bought(x, y)$
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- ① "Frank bought a dvd."
 - ② "Frank bought something."
 - ③ "Susan bought everything that Frank bought."
 - ④ "If Frank bought everything, so did Susan."
 - ⑤ "Everyone bought something."
 - ⑥ "Someone bought everything."

Expressing properties in FOL

Define an appropriate language and formalize the following sentences using FOL formulas.

- ① All Students are smart.
- ② There exists a student.
- ③ There exists a smart student.
- ④ Every student loves some student.
- ⑤ Every student loves some other student.
- ⑥ There is a student who is loved by every other student.
- ⑦ Bill is a student.
- ⑧ Bill takes either Analysis or Geometry (but not both).
- ⑨ Bill takes Analysis and Geometry.
- ⑩ Bill doesn't take Analysis.
- ⑪ No students love Bill.

Expressing properties in FOL

- ❶ $\forall x(Student(x) \rightarrow Smart(x))$
- ❷ $\exists x Student(x)$
- ❸ $\exists x(Student(x) \wedge Smart(x))$
- ❹ $\forall x(Student(x) \rightarrow \exists y(Student(y) \wedge Loves(x, y)))$
- ❺ $\forall x(Student(x) \rightarrow \exists y(Student(y) \wedge \neg(x = y) \wedge Loves(x, y)))$
- ❻ $\exists x(Student(x) \wedge \forall y(Student(y) \wedge \neg(x = y) \rightarrow Loves(y, x)))$
- ❼ $Student(Bill)$
- ❽ $Takes(Bill, Analysis) \leftrightarrow \neg Takes(Bill, Geometry)$
- ❾ $Takes(Bill, Analysis) \wedge Takes(Bill, Geometry)$
- ❿ $\neg Takes(Bill, Analysis)$
- ⓫ $\neg \exists x(Student(x) \wedge Loves(x, Bill))$

Expressing properties in FOL

For each property write a formula expressing the property, and for each formula write the property it formalises.

- Every Man is Mortal

$$\forall x(Man(x) \rightarrow Mortal(x))$$

- Every Dog has a Tail

$$\forall x(Dog(x) \rightarrow \exists y(PartOf(x, y) \wedge Tail(y)))$$

- There are two dogs

$$\exists x \exists y(Dog(x) \wedge Dog(y) \wedge x \neq y)$$

- Not every dog is white

$$\neg \forall x(Dog(x) \rightarrow White(x))$$

- $\exists x.Dog(x) \wedge \exists y.Dog(y)$

There is a dog

- $\forall x \forall y(Dog(x) \wedge Dog(y) \rightarrow x = y)$

There is at most one dog